

ENGINEERING BLUEPRINT

Frontend Specification Document

The Storyteller Portfolio

A premium, highly interactive full-stack personal website designed to showcase professional software engineering capabilities through a scroll-driven narrative interface.

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AUTHOR	Senior UI/UX Designer & Frontend Architect
TARGET AUDIENCE	Frontend Developers, UI/UX Designers, QA Engineers, Project Managers

1. Introduction

1.1 Project Overview

The **"Storyteller Portfolio"** is a premium, highly interactive full-stack personal website designed to showcase professional software engineering capabilities through a scroll-driven narrative interface. Moving away from rigid grid templates, the application presents technical case studies chronologically like a digital storybook. The user interface balances a soft, organic, human feel with sophisticated enterprise architectural depth.

1.2 Document Purpose

This document serves as the comprehensive engineering blueprint for the frontend application. It defines the design tokens, visual vocabulary, user experience guidelines, global state architecture, and integration patterns required to build a performant, accessible, and high-fidelity user experience.

2. Design System

2.1 Color Palette

The color landscape relies on a warm, low-contrast "Warm Minimalist" palette. It uses organic tones to create a premium editorial aesthetic that reduces eye strain while maintaining proper accessibility ratios.

Token / Variable Name	Hex Value	Role / Description
Primary Canvas Background	 #FAF8F5	Warm off-white / Light Alabaster canvas base
Primary Typography	 #2C2523	Deep Charcoal Espresso for contrast text and titles
Secondary Typography	 #60534E	Soft Clay Brown for muted body copy and captions
Subtle Accent Alpha	 #D4C9C1	Muted Warm Gray for "The Thread" timeline element
Success / Metric Highlight	 #7A8F75	Dusty Sage Green for statuses, numbers, and key metrics
Interactive / Focal Accent	 #C98A6B	Muted Terracotta / Soft Peach for primary buttons & highlights
Warm Highlight	 #E5A967	Warm Honey Yellow for Coffee/Chai micro-transaction call-to-actions

* Fallback Variables: Use standard placeholders `[SPECIFIC_COLOR_PALETTE]` for any runtime utility overrides.

2.2 Typography

Typography is paired to balance narrative elegance with clear technical documentation readouts.

- **Primary Heading Family (Display, H1, H2, H3):** Lora or Playfair Display (Fallback: Serif).
`[PRIMARY_FONT_FAMILY]` token maps here for editorial titles.
- **Secondary Body & Technical Text:** Plus Jakarta Sans or Inter (Fallback: Sans-Serif).
- **Monospace Text (Code Snippets, System Status):** JetBrains Mono or Fira Code (Fallback: Monospace).

Level	Size / Line Height	Weight / Style
Display 1	3.5rem (56px) / 1.2	Bold (Editorial Headings)
Heading 1	2.25rem (36px) / 1.3	SemiBold
Heading 2	1.75rem (28px) / 1.4	Medium
Body Regular	1.0rem (16px) / 1.6	Regular
Code/Caption	0.875rem (14px) / 1.5	Regular Monospace

2.3 Iconography

- **Library Specification:** `[ICON_SET_NAME]` (Defaulting to Lucide React / SVG components).
- **Stroke Width:** 1.75px uniform treatment.
- **Sizing Variants:** Small (16px), Medium (24px), Large (32px).

2.4 Spacing & Layout

A flexible, 8pt spatial grid system dictates padding, margins, and layout configurations.

- **Base Unit:** 8px (0.5rem).
- **Spacing Scale:** xs (4px), sm (8px), md (16px), lg (24px), xl (32px), xxl (48px), xxxl (64px).
- **Layout Wrapper:** max-width: 1280px.
- **Grid Configuration:** 12-column layout with 24px gutters for standard desktop views; flexible single-column layouts for narrative tracks.

2.5 Component Library (Overview)

All functional UI primitives are developed isolation-first within [SPECIFIC_DESIGN_TOOL_E.G._FIGMA_SKETCH] before implementation.

- **Buttons:** Standard padding 12px 24px. Rounded border-radius 12px (Soft bubble feel). Transitions set to all 0.3s cubic-bezier(0.25, 1, 0.5, 1).
- **Cards:** Zero harsh borders. Background: White with 40% opacity blur overlaying the canvas; box-shadow tracking: 0 4px 30px rgba(44, 37, 35, 0.03).

3. User Interface (UI) Mockups & Wireframes

3.1 Screen Layout Breakdown (Vertical Scroll Sequence)

◦ Storyteller Portfolio

● System Status: Live/Healthy

CHAPTER 1: THE PROLOGUE

"Hi, I'm a full-stack developer..."

[Hero Canvas Viewport]

CHAPTER 2: THE CHRONICLES (PIN-SCROLLING SPLIT VIEW)

[System Architecture Diagram Component]

SVG Canvas (Data Paths Pulse)

• [Spark] Inception Narrative

• [Blueprint] Systems Layer

• [Plot Twist] Challenges

• [Finale] Metrics & Output

CHAPTER 3: THE TOOLKIT

Backend Bookshelf

Java, Spring Boot, PostgreSQL

Frontend Bookshelf

TypeScript, React, Framer Motion

CHAPTER 4: THE EPILOGUE

[Postcard Contact Form]

☕ Gift a Cup Component

3.2 Key Component Specifications

- **The Narrative Thread Line:** A vertical, centered or left-aligned vector SVG element that tracks viewport scroll velocity. Using CSS masks or SVG path length strokes, it draws itself dynamically down the screen as the user scrolls.
- **Interactive Blueprint Modal:** A container block that transforms static project layouts into detailed system architecture topologies showing frontend-to-backend data layers.
- **The Postcard Contact Box:** Form designed with a subtle handwritten feel overlaying structured enterprise inputs.

4. User Experience (UX) Flow

4.1 Primary User Journey: The Chronicles

- **Entry:** User lands on the canvas. The user interface triggers a soft animation sequencing the text opacity from 0% to 100%.
- **Engagement:** As the reader scrolls vertically, the scrolling experience coordinates with micro-interactions. Elements do not snap abruptly; they use an ease-out timing curve to step smoothly into focus.
- **The Architecture Reveal:** The viewport docks on the Architecture step of a project. The window locks vertically (scroll-pinning), while horizontal text details cycle out on the right side and the SVG data paths inside the Architecture Diagram component pulse or animate data packets between boundaries.
- **Completion:** The scroll-lock releases once all project chapters are read, allowing the user to seamlessly proceed down to the interactive toolkit bookshelves.

4.2 Support Interaction: Micro-rewards

- **Bookshelf Hover:** Hovering over a specific framework card (e.g., Spring Boot) fires an action that dims unrelated timeline modules, drawing visual focus directly to the projects that implemented that specific technology.
- **The Coffee Interface:** Clicking  **Gift a Cup** dynamically converts the button area into an interactive payment gateway module without forcing an unprompted external tab redirection.

5. Frontend Architecture

5.1 Technology Stack

- **Core Framework Platform:** [KEY_FRAMEWORK_E.G._REACT_ANGULAR_VUE] (Standardized configuration built with TypeScript).
- **Animation Engine:** Framer Motion combined with custom scroll hooks.

- **Styling Engine:** Tailwind CSS or Styled Components mapping to structural design tokens.

5.2 Project Structure

The folder architecture follows a modular, feature-oriented structure:

```
src/
├── assets/           # Global styles, raw vectors, typography assets
├── components/       # Atomized common primitives (Buttons, Inputs, Modals)
│   └── ui/
├── features/         # Feature blocks driving modular logic
│   ├── archive/      # Project timeline components, architecture visualizers
│   ├── toolkit/       # Dynamic bookshelf components
│   └── transactional/ # Coffee support components, billing interaction handlers
├── hooks/            # Custom scroll-tracking and viewport observation hooks
├── routing/           # Configuration routes
├── services/          # Client API networks communicating with backend targets
└── store/             # Shared global UI state contexts
```

5.3 State Management

- **Engine Target:** `[STATE_MANAGEMENT_LIBRARY_E.G._REDUX_VUEX]` or React Context API.
- **Scope Responsibilities:** Coordinates global scrolling progression tracking, cached repository values fetched from the cloud backend, dark/light accessibility theme flags, and active state monitoring interfaces.

5.4 Routing

- **Engine Target:** `[ROUTING_LIBRARY_E.G._REACT_ROUTER]`.
- **Execution Profile:** Configured for single-page structural layout scrolling routes with deep-linking capability enabled (`/project/resume-analyzer`) to allow users to directly access specific system architecture chapters.

5.5 API Integration Strategy

- **Base Address Endpoint:** `[API_ENDPOINT_BASE_URL]`
- **Client Core Engine:** Axios instance layer featuring strict interception configurations.

Integrations Profile:

- `GET /api/v1/health` : Pings the running Java/PostgreSQL infrastructure runtime status to update the real-time footer metric pulse.
- `GET /api/v1/projects` : Hydrates data grids dynamically.
- `POST /api/v1/coffee/webhook` : Intercepts transaction updates securely from provider gateways.

5.6 Build & Deployment Process

The application utilizes an automated continuous delivery platform pipeline. Code check-ins generate automated production builds compiled down to static, globally cached assets optimized across edge networks (CDN targets like Vercel or Netlify).

6. Accessibility Standards (a11y)

- **Compliance Standard:** WCAG 2.1 AA Guidelines.
- **Contrast Targets:** All text elements must map safely above the 4.5:1 contrast ratio marker relative to the off-white custom background canvas.
- **Keyboard Operation:** Scroll-pinned phases must map safely to standard page navigation keystrokes (`ArrowDown` , `ArrowUp` , `Spacebar` , `PageDown` , `PageUp`). Focus rings must clearly outline fields during manual keyboard tracking sequences.
- **Screen Readers:** Use semantic HTML tags alongside descriptive `aria-live` regions to announce dynamic updates, such as when system architecture views shift or update on-screen.
- **Motion Controls:** Implement standard media query checks (`prefers-reduced-motion: reduce`). When active, scroll-pinned transformations must gracefully fall back to linear, static-block visibility configurations.

7. Performance Considerations

- **Lighthouse Target Matrix:** Performance ≥ 95 , Accessibility = 100, Best Practices = 100, SEO ≥ 95 .
- **Core Web Vitals:**
 - Largest Contentful Paint (LCP): Under 1.8 seconds.
 - Interaction to Next Paint (INP): Under 200 milliseconds.
 - Cumulative Layout Shift (CLS): Absolute target of 0.
- **Optimization Rules:** Image content must be rendered in modern formats (`.webp` or `.avif`) with explicit bounding dimensions. Lazy-load non-critical visual assets positioned outside the initial landing viewport.

8. Glossary

- **Scroll-Pinning:** A visual technique where a component locks vertically in the user's viewport during scrolling while other inner animation frames trigger horizontally or smoothly morph in place.
- **Design Tokens:** Centralized, platform-agnostic variables containing raw values for typography, colors, padding parameters, and animations.
- **System Architecture Visualizer:** A specialized UI widget designed to clearly show data flows and connections between frontends, APIs, and databases.

- **Micro-interactions:** Small, contextual user interface responses (like hover lifts, loading pulses, or steam effects) that provide elegant visual feedback.